How to Kernelize SVM

Example: **Polynomial Kernel**

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j$$ [ ]

$$W = \sum_{} \alpha_i y_i X_i$$

$$b = -\frac{1}{2}\left[\max_{i,\, y=-1} [\ ] + \min_{i,\, y=+1} [\ ] \right]$$

Class of vector **Z** ⇒ sign(W^T Z [ ] + b)

Kernelize using

$$\mathbf{K}(\alpha, b) = (\alpha^T b + 1)^d$$

$$\mathcal{L}_D(\alpha) = \sum_{i=1}^{N} \alpha_i - \frac{1}{2} \sum_{j=1}^{N} \sum_{i=1}^{N} \alpha_i \alpha_j y_i y_j$$ [ ]

$$W = \sum_{} \alpha_i y_i X_i$$

$$b = -\frac{1}{2}\left[\max_{i,\, y=-1} [\ ] + \min_{i,\, y=+1} [\ ] \right]$$

Class of vector **Z** is sign($[W^T Z + 1]^d$ [ ] + b)